

Quality attribute	Requirement	Design response
Maintainability	Domain changes should remain localized and database evolution reproducible	Modular package structure, controller–service–repository separation, DTOs, configuration properties, Flyway migrations, and automated tests
Usability	Users must understand loading, errors, empty results, score context, and available actions	Role-specific pages, normalized labels and reasons, validation signals, and explicit action controls
Auditability	Significant automated or administrative events should be reconstructable	audit_log, notification delivery records, email-action state, timestamps, actor/source metadata, and administrative views
Explainability	The system should expose evidence without presenting a score as a hiring probability	Lexical term/skill reasons, separate labels and application states, policy settings, and documented limitations

Security and auditability are defense-in-depth properties. A role rule at the URL layer does not replace ownership verification in a service, and an audit row does not make an unsafe action safe. Similarly, a database index is a performance design decision, not proof of latency under load. The evaluation chapter must test each critical claim in the environment in which it is reported.

1.5 Use-Case Analysis

The following fourteen use cases represent the main role-based goals available through the current frontend, actionable email pages, and protected backend workflows. Related screens are grouped into one user goal instead of treating every button or passive notification as a separate use case.

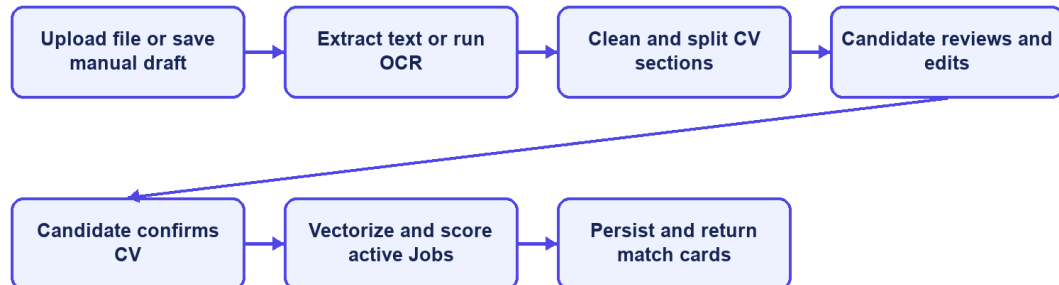
1.5.1 Candidate Manages CV, Profile, and Portfolio and Receives Match Results

Table 1.5. Use case - Candidate manages CV, profile, portfolio, and match results

Field	Description
Use-case ID	UC-01
Primary actor	Candidate
Preconditions	The Candidate account is active and authenticated. Candidate-owned profile, CV, and portfolio APIs are available.

Field	Description
Trigger	The Candidate uploads or drafts a CV, edits profile/portfolio data, manages an existing CV, or opens its match results.
Main flow	Create or upload a CV; validate and extract content; review and edit sections; confirm the CV; update profile and portfolio links/projects; list owned CVs; set an eligible SCORING_DONE CV as default or delete a permitted CV; vectorize confirmed content; score active Jobs; show the score, label, Potential explanation, and reasons.
Alternative/exception flows	Save incomplete manual content as DRAFT; keep extracted content in REVIEW_REQUIRED until confirmation; reject unsafe or unsupported files; record FAILED processing; reject a default change for an unscored CV; protect portfolio data according to the Candidate's privacy settings.
Postconditions	The owned profile, portfolio, and CV collection reflect the accepted changes, and confirmed CVs expose current matching results when scoring completes.

CV review and matching sequence



CareerFit IT AutoPilot — thesis diagram

Figure 1.5. CV review, confirmation, and matching sequence

1.5.2 Candidate Searches and Reviews Job Descriptions

Table 1.6. Use case - Candidate searches and reviews Job descriptions

Field	Description
Use-case ID	UC-02
Primary actor	Guest or Candidate

Field	Description
Preconditions	Public Job discovery requires no account. Personalized information and recruitment actions require an authenticated Candidate.
Trigger	The user searches, applies filters, selects an urgent or suggested Job, or opens a Job/employer detail page.
Main flow	Send server-side search, filter, sort, and pagination parameters; display active Jobs; open the selected JD; show structured Job and employer information; load similar Jobs; retain the selected filters when the user returns to the catalogue.
Alternative/exception flows	Show a clear empty state when no Job matches; show a retryable error when the API fails; do not expose closed, hidden, banned, or otherwise ineligible Jobs through the public catalogue.
Postconditions	The user receives current Job information without creating an Application; authentication is requested before a protected action.

1.5.3 Candidate Manages Applications and Recruiter Invitations

Table 1.7. Use case - Candidate manages applications and Recruiter invitations

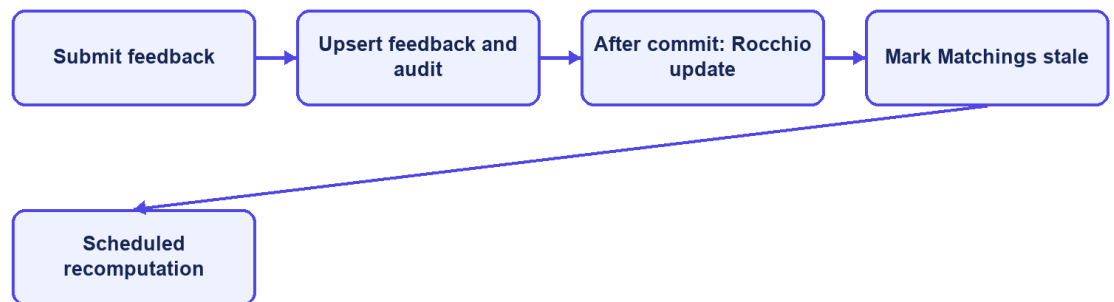
Field	Description
Use-case ID	UC-03
Primary actor	Candidate
Preconditions	The Candidate is authenticated. Applying requires an active Job before its deadline and a confirmed selected or default CV.
Trigger	The Candidate applies, opens application history, withdraws an application, or accepts/declines a Recruiter invitation.
Main flow	Validate the Candidate, Job, deadline, CV, and duplicate state; create the Application with available Matching context; list and filter owned Applications; withdraw when the state permits; display pending invitations; accept or decline an invitation; notify the related parties and record important changes.
Alternative/exception flows	Reject missing authentication/CV, duplicate application, inactive or expired Job, another Candidate's Application, or an unsupported withdrawal/invitation transition.
Postconditions	The Application or invitation response is stored with a valid lifecycle state and remains visible in the Candidate's history.

1.5.4 Candidate Provides Match Feedback and Updates Later Ranking

Table 1.8. Use case - Candidate provides match feedback and updates later ranking

Field	Description
Use-case ID	UC-04
Primary actor	Candidate
Preconditions	The Candidate owns the CV and Matching connected to the feedback.
Trigger	The Candidate submits GOOD_MATCH, POTENTIAL, BAD_MATCH, or NOT_INTERESTED.
Main flow	Resolve the Matching and authenticated actor; verify ownership; upsert one current judgment; store audit evidence; commit the transaction; start Rocchio learning for supported positive/negative feedback; mark affected Matchings for recomputation; rescore and clear the stale marker after success.
Alternative/exception flows	Reject another role, another Candidate's Matching, missing evidence, or invalid feedback; store NOT_INTERESTED without treating it as a technical relevance signal; retain retry visibility when asynchronous recomputation fails.
Postconditions	The feedback is stored, and later rankings use the rebuilt learned Job vector when recomputation completes.

Feedback recomputation sequence



CareerFit IT AutoPilot — thesis diagram

Figure 1.6. Feedback learning and recomputation sequence

1.5.5 Candidate Reviews Personalized Recommendations and Analytics

Table 1.9. Use case - Candidate reviews personalized recommendations and analytics

Field	Description
Use-case ID	UC-05
Primary actor	Candidate
Preconditions	The Candidate is authenticated. Personalized results require available profile, preference, CV, Matching, or analytics data.
Trigger	The Candidate opens Recommendations or Advanced Analytics.
Main flow	Load personalized Job recommendations separately from the Matching feed; open, skip, or apply from a recommendation; load matching trends, skill demand, and profile-gap summaries; present the results as decision support; allow the Candidate to return to profile/CV management when improvement is needed.
Alternative/exception flows	Show an empty or limited-data state when the profile, confirmed CV, or analytics history is insufficient; show a retry action for API failure; do not create an Application only because a Job was recommended.
Postconditions	The Candidate can use the recommendation and analytics evidence to choose a Job or improve the profile without an automatic recruitment decision.

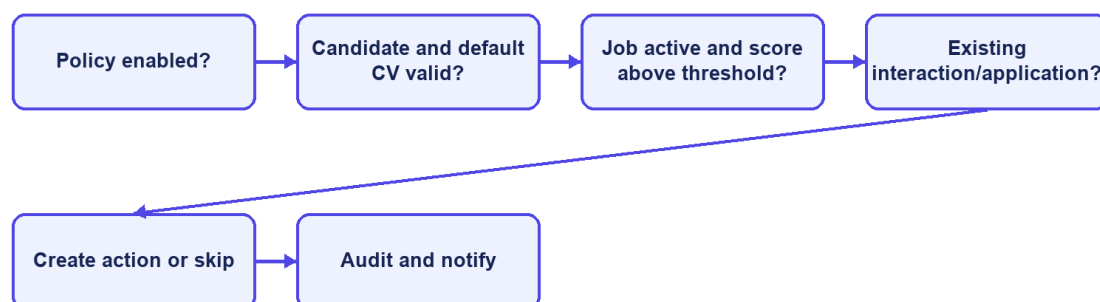
1.5.6 Candidate Configures and Runs AutoFit

Table 1.10. Use case - Candidate configures and runs AutoFit

Field	Description
Use-case ID	UC-06
Primary actor	Candidate and background scheduler
Preconditions	The Candidate owns an AutomationPolicy. Execution also requires an eligible confirmed default CV and completed matching data.
Trigger	The Candidate changes AutoFit settings, pauses/resumes automation, selects Run Now, or a scheduled scan evaluates the policy.
Main flow	Save enablement, thresholds, categories, quota, cooldown, time zone, quiet hours, and notification preferences; resolve the Candidate, CV, Jobs, Matchings, consent, and previous interactions; select only eligible actions; create permitted Applications or notifications; write delivery and audit evidence.

Field	Description
Alternative/exception flows	Skip disabled, paused, ineligible, duplicate, stale, quota-exceeded, cooldown, quiet-hour, invalid-deadline, or invalid-state items; isolate a failed item instead of stopping the full scan.
Postconditions	Only policy-allowed actions are recorded; skipped items remain unchanged and explainable through stored outcomes.

AutoFit decision flow



CareerFit IT AutoPilot — thesis diagram

Figure 1.7. AutoFit decision flow

1.5.7 Candidate or Recruiter Reports Suspicious Recruitment Content

Table 1.11. Use case - Candidate or Recruiter reports suspicious recruitment content

Field	Description
Use-case ID	UC-07
Primary actor	Candidate or Recruiter
Preconditions	A Candidate may report an ACTIVE Job. A Recruiter may report a CV only through an owned Job that provides visibility by Application or Matching.
Trigger	The actor opens Report on an eligible Job or CV, selects a supported reason, and optionally enters a comment.
Main flow	Resolve the actor and target; apply the role, Job ownership, CV visibility, and target-state checks; validate the reason/comment; prevent another pending report from the same reporter for the same target; create the PENDING ContentReport; increase the target counter; add audit evidence; expose the case to Administrator moderation.

Field	Description
Alternative/exception flows	Require a comment for OTHER; reject an inactive Job, banned CV, unrelated Recruiter, invisible CV, unsupported reason, or duplicate pending report.
Postconditions	The target has a pending report count, and the report appears in the Administrator queue until it is dismissed or actioned.

1.5.8 Recruiter Manages Company Profile and Job Lifecycle

Table 1.12. Use case - Recruiter manages company profile and Job lifecycle

Field	Description
Use-case ID	UC-08
Primary actor	Recruiter
Preconditions	The Recruiter account is active and authenticated. A company profile is required before publishing a Job.
Trigger	The Recruiter completes onboarding, edits the company profile, creates/edits a Job, changes its status, deletes it, or exports the owned Job list.
Main flow	Create or update the company profile; list owned Jobs; enter structured and free-text JD data; preview quality signals; save DRAFT; review and publish; set urgent and deadline fields; edit an owned Job; change among supported Recruiter lifecycle states; delete a permitted Job or export the list; start vectorization and scoring after a valid publish commit.
Alternative/exception flows	Reject missing company data, blocking JD/deadline errors, another Recruiter's Job, an unsupported transition, or deletion that violates current state rules; retain incomplete work as DRAFT.
Postconditions	The company and owned Job state reflect the accepted operation, and published eligible Jobs are available for matching and applications.

1.5.9 Recruiter Reviews and Processes Applicants

Table 1.13. Use case - Recruiter reviews and processes applicants

Field	Description
Use-case ID	UC-09
Primary actor	Recruiter
Preconditions	The Recruiter is authenticated and owns the selected Job.
Trigger	The Recruiter opens the ranking/applicants workspace or chooses an application decision.

Field	Description
Main flow	Load and filter applicants by valid states; show Candidate, CV, Matching, label, Potential, and permitted portfolio information; open a Candidate detail; approve or reject an eligible Application; update the list; send configured lifecycle notifications; add audit evidence.
Alternative/exception flows	Reject a Job owned by another Recruiter, an unknown Application, hidden private data, or an invalid status transition; show empty states for Jobs without applicants.
Postconditions	The Application contains the valid Recruiter decision and the Candidate can observe the updated state.

1.5.10 Recruiter Manages Talent Pool and Candidate Invitations

Table 1.14. Use case - Recruiter manages Talent Pool and Candidate invitations

Field	Description
Use-case ID	UC-10
Primary actor	Recruiter
Preconditions	The Recruiter is authenticated and owns the Job used for Candidate discovery.
Trigger	The Recruiter opens Discover, Bookmarked CVs, or Invited CVs for an owned Job.
Main flow	Load matching and Potential Candidates with filters; view permitted CV and explanation data; bookmark or unbookmark a Candidate for the Job; send an invitation; list invitations awaiting a response; withdraw a pending invitation when permitted; update the corresponding tabs and notifications.
Alternative/exception flows	Reject unrelated Job ownership, an invisible/missing Candidate, duplicate invitation, invalid invitation withdrawal, or an unsupported Candidate state; show a clear empty state for each tab.
Postconditions	Bookmark and invitation records reflect the Recruiter's choices and remain scoped to the owned Job.

1.5.11 Recruiter Reviews Analytics and Configures Recruitment Notifications

Table 1.15. Use case - Recruiter reviews analytics and configures recruitment notifications

Field	Description
Use-case ID	UC-11
Primary actor	Recruiter
Preconditions	The Recruiter is authenticated; private recruitment analytics depend on available owned-Job activity.
Trigger	The Recruiter opens Analytics/Advanced Analytics or recruitment notification settings.

Field	Description
Main flow	Load recruitment overview and trends; show owned-Job/application outcomes and market, skill, or salary summaries with their scope; enable or disable high-match CV alerts, new-application alerts, invitation-response alerts, approval digests, and Job-closing reminders; set the high-match threshold; save the preferences.
Alternative/exception flows	Show limited-data or empty states when activity is insufficient; distinguish platform-wide market data from private company results; retain the previous settings if an update fails.
Postconditions	The Recruiter can monitor current evidence, and later notification decisions use the stored preferences.

1.5.12 Administrator Manages User Accounts and Job Visibility

Table 1.16. Use case - Administrator manages user accounts and Job visibility

Field	Description
Use-case ID	UC-12
Primary actor	Administrator
Preconditions	The Administrator is authenticated and authorized for administrative APIs.
Trigger	The Administrator searches users/Jobs and selects Suspend, Activate, Hide, or Restore.
Main flow	Load paginated user or Job data; apply filters; suspend an ACTIVE account or activate a suspended account; hide an ACTIVE Job using HIDDEN_BY_ADMIN; restore only a Job hidden by Administrator; refresh the result; record the administrative action and target in audit history.
Alternative/exception flows	Reject missing targets, invalid roles, or unsupported state changes; never use Restore to publish a DRAFT, PAUSED, CLOSED, or BANNED Job.
Postconditions	The selected account state or Job visibility is updated without changing unrelated recruitment records.

1.5.13 Administrator Reviews and Resolves Content Reports

Table 1.17. Use case - Administrator reviews and resolves content reports

Field	Description
Use-case ID	UC-13
Primary actor	Administrator
Preconditions	The Administrator is authenticated. At least one report case may be pending for a JOB or CV target.
Trigger	The Administrator opens Report moderation, selects a queue/case, and chooses Dismiss or Ban.

Field	Description
Main flow	Load grouped pending JOB/CV cases; show target details, reasons, comments, reporters, and timestamps; lock and reload the target; dismiss all pending reports when no action is needed or set the target to BANNED when action is required; clear the target counter; resolve the pending report records; add audit evidence; refresh the queue.
Alternative/exception flows	Show an empty queue when no case is pending; reject a missing target, a case with no pending reports, or an invalid target type; keep the case unchanged if the transaction fails.
Postconditions	Pending reports become DISMISSED or ACTIONED, and a banned target is excluded from normal use according to its domain rules.

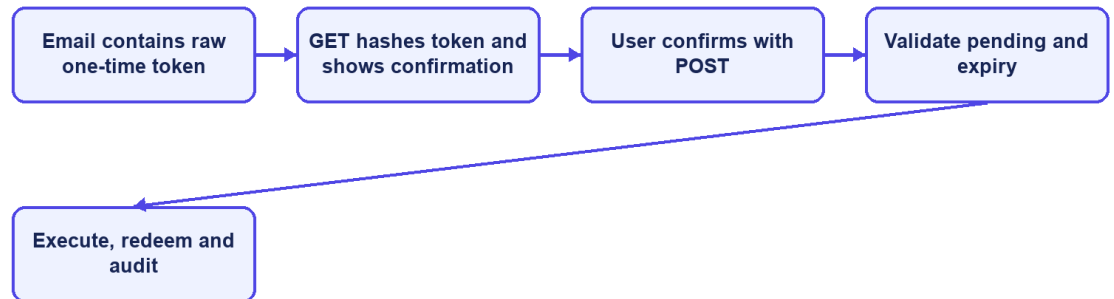
1.5.14 Recipient Reviews and Completes an Actionable Email

Table 1.18. Use case - Recipient reviews and completes an actionable email

Field	Description
Use-case ID	UC-14
Primary actor	Email recipient
Preconditions	A pending, unexpired EmailAction exists for the recipient, and only the SHA-256 token hash is stored.
Trigger	The recipient opens an action from MATCH_NOTIFICATION, DAILY_DIGEST, or RECRUITER_INVITATION and deliberately confirms it.
Main flow	GET hashes and validates the token and shows a non-mutating confirmation page; POST repeats validation and performs the selected action. Match notification supports GOOD_MATCH, POTENTIAL, or NOT_INTERESTED; Daily Digest supports per-Job feedback and UNSUBSCRIBE_DIGEST; Recruiter Invitation supports INVITATION_ACCEPT or INVITATION_DECLINE. The service records redemption and audit/feedback/application effects.
Alternative/exception flows	Reject missing, malformed, expired, redeemed, or unsupported tokens/actions; do not change recruitment state from GET; show a clear result for unsuccessful redemption.
Postconditions	The requested action is executed at most once under normal validation. Other one-way lifecycle emails remain notifications rather than separate use cases.

Other lifecycle emails are treated as notifications rather than separate use cases because they inform the recipient without adding another token-based user goal or a separate state-changing workflow.

Hardened email-action sequence



CareerFit IT AutoPilot — thesis diagram

Figure 1.8. Secure email-action confirmation and redemption sequence

1.6 Chapter Summary

This chapter defined CareerFit's actors, functional and non-functional requirements, and fourteen representative use cases. The use cases follow real role-based goals across the frontend, actionable email flow, and protected services. They keep matching evidence, application state, user policy, report moderation, automated actions, and audit records separate so later design and implementation decisions can be traced to expected behavior.

CHAPTER 2. THEORETICAL BACKGROUND AND SOLUTION DESIGN

2.1 Recruitment Information and CV–Job Description Matching

Recruitment matching is an information-retrieval and decision-support problem that compares different types of evidence about candidates and vacancies. A curriculum vitae usually describes education, work history, projects, technical skills, certifications, languages, and contact information. A job description lists responsibilities, required and preferred skills, seniority, location, employment conditions, and company context. Some data is structured, such as years of experience or location, while other data is free text and may use inconsistent terms. Therefore, exact database filters are useful for fixed conditions but are not enough to rank the overall fit between a candidate and a position.

The same information supports different user tasks. From the candidate perspective, the system retrieves and ranks jobs that fit a CV or desired profile. From the recruiter perspective, it ranks applicants or discovers potential candidates for a specific vacancy. These directions share document representations and similarity functions, but they are not identical business decisions. A relevant job recommendation does not imply that an application has been submitted, and a high candidate–job similarity does not imply that a recruiter should accept the candidate. The score is therefore evidence for prioritization rather than a final employment decision.

Recruitment data also has domain-specific ambiguity. A technology may be expressed by an abbreviation, a product name, or a broader skill family; job titles vary across companies; and the same term may have different importance at junior and senior levels. Open recruitment corpora are relatively uncommon because CVs contain personal information. The Djinni Recruitment Dataset illustrates both the scale of the matching problem and the value of anonymized, documented corpora, providing more than 150,000 jobs and 230,000 candidate profiles in English and Ukrainian [6]. CareerFit is narrower: it focuses on IT recruitment and uses explicit fields and normalized text so that matching behavior can be inspected within the project scope.